

Chuxuan Li

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EDUCATION

University of Pennsylvania

Ph.D. candidate, Bioinformatics

GPA: 3.9; **Advisor:** Dr. Mingyao Li

Philadelphia, PA

Expected June 2028

The University of Chicago

B.A. *summa cum laude*, Biological Sciences and Psychology

GPA: 3.934; Biological Sciences **Major GPA:** 3.963

Relevant courses: Mathematical Methods for Biological Sciences I-II, Linear Algebra, Computer Science with Applications I-II-III, Proteomics and Genomics in Molecular Engineering, Fundamentals of Genetics

Chicago, IL

June 2021

RESEARCH EXPERIENCE

Department of Psychiatry and Behavioral Neuroscience, **The University of Chicago**

Research Assistant

Evanston, IL

August 2021-May 2023

Advisors: Dr. Jubao Duan, Dr. Xin He, Dr. Siwei Zhang

Project: Activity-dependent regulation of neuropsychiatric risk through cell-type specific chromatin landscape changes.

- Designed and implemented pipeline for single-cell multiomic (scRNA-seq and scATAC-seq) quality control and downstream analysis;
- Found differentially expressed (DE) genes, differentially accessible (dynamic) peaks, and allele-specific open chromatin (ASoC) in response to neural stimulation;
- Performed enrichment test of neuropsychiatric risk loci in DE genes and dynamic peaks;
- Presented posters at the 2022 meeting of the American Society of Human Genetics; co-first author for manuscripts under preparation.

Biological Sciences Department, **The University of Chicago**

Research Assistant

Chicago, IL

March 2020-June 2021

Advisor: Dr. Sebastain Pott

Project: Characterization of the human heart cell atlas with scRNA-seq and scATAC-seq.

- Performed QC and label transfer on single-cell RNA-seq and ATAC-seq data using mainly Cell Ranger, Seurat, and ArchR; found subtypes of cardiomyocytes marked by distinct gene expression profiles;
- Found novel gene groups from single-cell RNA-seq data on human heart cells using topic modeling.

RESEARCH INTERESTS

High-dimensional genetic data integration, machine learning, single-cell multiomic analysis, gene co-expression networks, functional interpretation of chromatin structural variant.

TEACHING EXPERIENCE

Biological Sciences Department, **The University of Chicago**

Teaching Assistant

Chicago, IL

March 2020-June 2021

Courses: Mathematical Methods for Biological Sciences, Introduction to Quantitative Modeling in Biology

- Led weekly computational lab sessions and office hours for around 15 students per course;
- Advised students on final projects and graded all assignments and projects.

PRESENTATIONS AND ABSTRACTS

Li, C. et al. (2022, October 25-29). *Cell-type-specific activity-dependent multiomic profiling in single hiPSC-derived neurons for neuropsychiatric disorders*. [Poster presentation]. American Society of Human Genetics Annual Meeting, Los Angeles, CA, United States.

Li, C. et al. (2022, October 25-29). *Cellular context-specific gene regulation of neuropsychiatric disorders in single human neurons*. [Poster presentation]. American Society of Human Genetics Annual Meeting, Los Angeles, CA, United States.

Zhang, S., Zhang, H., Liang L., Li, C., Jamison, B., Wood, W., Kozlova, A., Sanders, A., Pang, Z., He, X., Duan, J. (2022, September 13-17). *Cellular context-specific gene regulation of neuropsychiatric disorders in single human neurons*. [Abstract]. World Congress of Psychiatric Genetics. Florence, Italy.

MANUSCRIPT UNDER PREPARATION

Context-specific epigenetic and genetic regulations of neuropsychiatric disorders at single neuron resolution. Co-first author. Presented at the American Society of Human Genetics Annual Meeting 2022.

HONORS AND PROFESSIONAL MEMBERSHIP

Member of the American Society of Human Genetics 2021, 2022

Phi Beta Kappa 2020

Dean's List 2019, 2020, 2021

Awarded to the top 20% of degree-seeking students for that academic year.

Gary Becker Scholars 2020, 2021

Awarded to the top 5% of students of their major relative to their peers in the major over a five-year period.

SERVICE AND LEADERSHIP

Emergency Department, **The University of Chicago Medicine** Chicago, IL
Medical Scribe January 2021-June 2021

- Maintained records of patient visits, briefed physicians on patient history before visits.

TechTeam, **The University of Chicago** Chicago, IL
Student Researcher October 2019-March 2020

- Conducted user interface and experience research with a team of ten students on the accessibility of the official website for UChicago Student Wellness;
 - Co-designed a new layout and a chatbot function to facilitate easy navigation of the website.

SKILLS

Programming: Proficient in **Python** (Pandas, Beautiful Soup) and **R** (Seurat, Signac, ArchR, tidyverse, Limma, EdgeR, DESeq2), working knowledge in **Linux**.

Experimental: Cell cloning, PCR, gel electrophoresis, CRISPR editing, cell culture, NMR, western blot.

Languages: Mandarin Chinese (Native), English (Advanced), Japanese (Intermediate).